

Ahmed El-Sayed Massoud

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Education

Alexandria University

Master of Science (MSc) in Molecular Biology.

Alexandria, Egypt
Sep 2022 – Aug 2026

- Coursework Year CGPA: **3.93/4.00** | Grade: **Excellent (98.25%)**.
- Published a research article in a Q1-ranked, peer-reviewed journal.
- Master's Thesis: *Analgesic Activity of Betanin on a Rat Model of Inflammatory Pain Induced by Complete Freund's Adjuvant*.

Alexandria University

Bachelor of Science (BSc) in Molecular Biology and Microbiology.

Alexandria, Egypt
Sep 2015 – Aug 2019

- CGPA: **3.69/4.00** | Grade: **Excellent (90.70%)**.
- Graduated as the 1st-ranked student in the Molecular Biology major.
- Research Project: *Gene Therapy Approaches for Type 1 Diabetes Mellitus (T1DM)*.

Research Contributions (Publications & Scholarly Activities)

Publications

	Title	Journal
1	Zayed, M., Massoud, A. , Hassan, F., Elwakeel, E., Mahmoud, Y., & Jeong, B.-H. (2025). Harnessing adipose ferroptosis: A promising novel pathway for obesity treatment. <i>Current Obesity Reports</i> , 14(1), 74. https://doi.org/10.1007/s13679-025-00669-1	Current Obesity Reports - IF: 11
2	Zayed, M., Massoud, A. , Al-Haddad, R. et al. Integrating Single-Cell Sequencing and Proteomics to Unravel Stem Cell Heterogeneity: Implications for Precision Regenerative Medicine. <i>Stem Cell Rev and Rep</i> (2026). https://doi.org/10.1007/s12015-026-11116-6	Stem Cell Reviews and Reports - IF: 4.2
3	Massoud, A. , Essawy, A. E., Alfreddan, M. A., Abdel-Moneim, A. M., Gomaa, R. A., & Abdel Salam, S. (2026). Betanin from <i>Beta vulgaris</i> Attenuates Complete Freund's Adjuvant-Induced Inflammatory Pain: Integrated Preclinical and In Silico Insights. <i>Biomedicines</i> , 14(6), 1202. https://doi.org/10.3390/biomedicines14061202	Biomedicines - IF: 3.9
4	Zayed, M., Massoud, A. , Elwakeel, E. et al. Repurposing approved drugs as ferroptosis modulators: a critical review of clinical trials in cancer and neurodegeneration. <i>Naunyn-Schmiedeberg's Arch Pharmacol</i> (2026). https://doi.org/10.1007/s00210-026-05582-5	Naunyn-Schmiedeberg's Archives of Pharmacology - IF: 4.0
5	Essawy, A. E., Fahmy, R. I., Gomaa, R. A., Fetouh, H. A., Helal, M., Massoud, A. , Hassan, M. A. & Abdel-Wahab, W. M. Neuroprotective effects of Moringa oleifera leaf extract via potential NF-κB modulation in cuprizone-induced demyelination in rats: Integrated in vivo and in silico evidence. <i>Tissue & Cell</i> (2026). https://doi.org/10.1016/j.tice.2026.103938	Tissue & Cell - IF: 3.1
6	Zayed, M., Mahmoud, M., Massoud, A. et al. The SLC7A11 Thermostat: A Molecular Signaling Switch Between Ferroptosis and Disulfidptosis in Neurodegenerative Disease. <i>Mol Neurobiol</i> 63, 916 (2026). https://doi.org/10.1007/s12035-026-06219-7	Molecular Neurobiology - IF: 5.5
7	Engineered MSC-derived extracellular vesicles for mitochondrial precision medicine in neurodegenerative diseases	Under review
8	Mesenchymal stem cell exosomes as architects of trained immunity: Epigenetic reprogramming of innate immune memory for disease modulation	Under review
9	Ferroptosis as a Therapeutic Target in Cardiovascular, Respiratory, and Renal Diseases: Repositioned Drugs and Translational Challenges	Under review

Poster Presentations

The City of Scientific Research and Technological Applications - SRTA City

Borg Al-Arab, Alexandria
Jul 2019

- Participated in the design and presentation of a scientific poster titled 'C.A.S.E. Study: Engineered Stem Cells Combating Cancer', focusing on the therapeutic potential of reprogramming patient-derived stem cells to target cancer.

Peer Review Activities

- Contributed to the peer-review process of Journal of Molecular Histology as an invited reviewer. Jul 2026
- Contributed to the peer-review process of Open Veterinary Journal as an invited reviewer. Nov 2025

Internships & Grants

Academy of Scientific Research and Technology - ASRT

Cairo, Egypt

ASRT Master's Scholar

Sep 2022 – Oct 2023

- Recipient of the prestigious Scientists for Next Generation (SNG) Program scholarship, 8th batch, awarded by ASRT based on academic merit for a master's degree.

Egypt Scholars Incorporation

Remote Learning

Research Intern

Jan 2023 - Nov 2023

- Completed ES Basic Lab 10, a structured course teaching fundamental scientific research skills and technical tools through recorded and interactive lectures.
- Collaborated in a team of six to design and present a project titled "Antibiotic Resistance & Global Warming", applying learned concepts and enhancing teamwork skills.
- Graduated with excellence from the Basic Lab, qualifying for ES Advanced Lab 10, a research internship focused on the practical application of research skills, including producing and publishing scientific articles.

Tawakkol Karman International Foundation - TKF

Remote Learning

Intern

Apr 2022 – Jun 2023

- The internship focused on empowering youth within the 'Leaders for Future Program', conducted by TKF.
- The program included three diplomas in 'Peace Building and Human Rights', 'Labor Market Preparation', and an 'Intensive English Language Diploma', as well as a preparatory course for TOEFL or IELTS exams in its final stage, delivered by English Castle Institute in Istanbul, Turkey.

Medical Biotechnology Department at SRTA City

Borg Al-Arab, Alexandria

Research Intern

Jul 2019 – Jul 2019

- The internship focused on "The Effect of Some Pharmaceutical & Natural Products as Anti-cancer Agents", gaining hands-on experience with techniques such as PCR, electrophoresis, flow cytometry, and tissue culture.

Pharmaceutical & Fermentation Industries Development Center at SRTA City

Borg Al-Arab, Alexandria

Research Intern

Jan 2019 – Feb 2019

- The internship focused on "Basic Tissue Culture & Molecular Biology Techniques" and involved hands-on experience with techniques such as Western blotting, PCR, ELISA, and tissue culture.

Professional Experience

Alamein International University

New Alamein, Matrouh

Full-time Assistant Lecturer

Oct 2023 - Present

Full-time Teaching Assistant

Oct 2023 - Sep 2023

- Full-time Teaching Assistant at the Molecular Biotechnology Program, Field of Advanced Basic Sciences.
University Profile: <https://aiu.edu.eg/profile/ahmed-elsayed-massoud>

- Assist in teaching basic biology, molecular biology, and biotechnology courses, including
 - Biology I (Anatomy and Cell Biology)
 - Biology II (Histology and Physiology)
 - Introduction to Biotechnology
 - Cell & Molecular Biology (for the Biomedical Engineering Program, Faculty of Engineering)
 - Vertebrates
 - Cellular Structure & Cell Communication
 - Bioinformatics
 - Cell Culture Principles & Techniques
 - Molecular Biology I
 - Molecular Biology II
 - Graduation Project
 - Recombinant DNA Technology
- Assist in preparing course assignments and labs, and answer keys for assignments under the guidance of the course instructor, in addition to exam proctoring and attendance recording.
- Supervise and assess students during the “Practical Training and Internship” graduation requirement.
- Help in managing course LMS contents and interact with students using discussion groups and forums on the LMS and via other communication means.
- Member of the “Academic Advising,” “Quality Assurance & Accreditation,” “Devices & Labs Preparation,” “Exams Supervision & Control,” “Reporting & Events Documentation,” and “Coursera” committees.
- Participate in the preparation of quality assurance and accreditation files.
- Fully responsible for student academic advising, event preparation and faculty logistics, in addition to leading administrative tasks, meetings, and conferences in my role as the senior teaching assistant.
- Hold office hours (face-to-face and online) to support and guide students.

El-Safwa Medical Laboratory Medical Laboratory Specialist

Amria City, Alexandria
Aug 2019 – Dec 2021

- Collect, handle, and prepare samples and reagents for analysis.
- Report needs for chemicals and glassware to ensure continuous availability.
- Perform clinical biochemical tests (liver profile, kidney profile, heart profile, lipid profile, etc.).
- Perform hematological tests (CBC manual and automated, ABO grouping, BT, CT, etc.).
- Perform serological and microbiological tests (CRP, RF, microbial culturing, and antibiotic assays).
- Carry out urine and stool tests investigating for pus, RBCs, salts, and parasites. Write down the results.
- Carry out qualitative & quantitative *H. pylori* (Ag, Ab), *Typhi/Paratyphi* (Ag, Ab), drug abuse tests, HCG, virus tests, and others.
- Operate instruments such as a spectrophotometer, centrifuge, microscope, pipettes, and ELISA reader.

Skills

Technical Skills & Laboratory Experience:

- Molecular & Cellular Techniques:** Tissue culture, nucleic acid and protein extraction, purity assessment, conventional PCR and qPCR, gel electrophoresis, and bacterial transformation.
- Biochemical Assays:** ELISA, colorimetric assessment of oxidative stress, and antioxidant enzyme assays.
- Neurobehavioral & Animal Studies:** Rotarod and hotplate testing; animal dissection (rat, fish, lizard, pigeon, toad); spinal cord extraction via hydraulic extrusion.
- Microbiology Techniques:** Media preparation, bacterial and fungal culturing, antibiotic sensitivity testing, Gram staining, and acid-fast staining.

Computer & Bioinformatics Skills:

- Office & Design Tools:** MS Word, PowerPoint, Excel, and Teams; Canva, Picsart, Adobe Photoshop, BioRender, and Canvas LMS.
- Reference & Research Management Tools:** EndNote, Mendeley, and Lean Library Workspace (Sciwheel).
- Scientific & Computational Tools:** ImageJ, SPSS; molecular docking and visualization (AutoDock, BIOVIA Discovery Studio, ChimeraX, PyMOL); chemical structure drawing (ChemSketch); ADME and toxicity profiling (SwissADME, ProTox); protein modeling (SWISS-MODEL); sequence and domain analysis (CDD, ScanProsite); and standard bioinformatics resources (PubChem, NCBI, EBI, SIB, PDB).

- **Specialized Skills:** Scientific illustration and publication-quality figure design | [View Portfolio](#)

Language: Arabic (native), English (TOEFL ITP - B2), French (A1), and German (A1).

Courses, Workshops & Conferences

Courses

- Diploma in Analytical Techniques in Purification of Biomolecules - Alison 2024
- Introduction on Nanotechnology - Edraak 2023
- Science Communication - Udemey 2022
- SPSS Statistics Essential Training - LinkedIn Learning 2022
- Excel Essential Training (Office365/Microsoft 365) - LinkedIn Learning 2022
- Basic First Aid skills – Edraak 2022
- Editing and Proofreading Made Simple - LinkedIn Learning 2022
- Writing with Proper Punctuation - LinkedIn Learning 2022
- Emotional Intelligence Basics – LinkedIn Learning 2022
- Speaking Confidently and Effectively - LinkedIn Learning 2022
- Critical Thinking and Problem Solving - LinkedIn Learning 2022
- Introduction to Scientific Writing - Education Clinic Inc. 2022
- Basics of Scientific Research – Egypt Scholars Inc. 2019
- Computer basics and MS Office - Alexandria University 2018

Workshops

- Western Blot Practical Techniques, Achieving Reliable and Reproducible Results Workshop – By Agitech & Bio-Rad at SRTA City 2024
- Finding Funding and Writing a Winning Proposal: Structure and Tips for Writing – TICO AIU 2023
- How to Deliver Inspiring Presentations - CABI 2022
- How to Craft Captivating Slides - CABI 2022
- How to Plan Thoughtful Oral Presentations - CABI 2022
- Structuring your article correctly - Researcher Academy | Elsevier 2022
- How to find the protocols and methods you need for your research – Springer Nature 2022
- Common Mistakes in Manuscript Writing - IET 2022
- How to Write Professional Response to Reviewers - CABI 2022
- How to Write Professional Review Report - CABI 2022
- How to review your manuscript before submission - CABI 2022
- How to write Conclusions, Abstract, Title & References - CABI 2022
- How to write Results and Discussions - CABI 2022
- How to write the Introduction and Materials & Methods - CABI 2022
- Entrepreneurship Workshop in the City of Scientific Research and Technological Applications 2019
- Scientific Research Workshop - Industrial Microbiology and Applied Chemistry Volunteers 2017
- Soft Skills Workshop - European Association of Geoscientists and Engineers 2016
- Leadership skills - Tamdeen for Training and Management Consulting 2015

Conferences

- 3rd Scientific Conference: Emerging Technologies and Their Relationship to Sustainable Development - City of Scientific Research and Technological Applications, Borg Al-Arab City. 2025
- Annual Arab Asian Biotechnology Conference (AABC 2025): Desert Biotechnology and Sustaining Life in Arid Regions - Bibliotheca Alexandrina, Egypt 2025
- 12th STUDENT & 5th POST-GRADUATE Conference for Research & Innovation - Faculty of Science, Alexandria University 2025
- The 9th Alexandria SBS Bowel Diseases: The Art and Science in Clinic and Research - Faculty of Medicine, Alexandria University 2024
- The 2nd International Conference of the City of Scientific Research and Technological Applications 2024
- First International Conference of the Egyptian Society of Biological and Biomedical Sciences 2024
- Young Pharmaceutical Scientists Association - YPSA Online Conference on 23rd to 28th of July 2022

Extracurricular Activities

Undergraduate Research Initiative (URI)

Online
Sep 2025

- Delivered a volunteer lecture on scientific literature searching, critical reading of research papers, and reference management tools during the Sci. Green Camp, organized by the URI, Alexandria University.
Lecture Video: <https://youtu.be/BVAsf40ujD0>

Health Promotion Week - AIU

Oct 2023 - Present

- Team Leader of Health Promotion Week, Faculty of Science, Alamein International University. Leading and supervising multiple student teams, coordinating logistics, and ensuring the smooth execution of educational and outreach activities focused on personal and environmental health awareness.

American Society for Microbiology (ASM)

Oct 2022 - Present

- Active member of the American Society for Microbiology. Membership ID: 200334234.

Mendeley, Elsevier

Jul 2022 - Present

- Registered as a Mendeley Advisor to share reference management skills and host training sessions.

Scholarjobs

Online platform
Feb 2022 - Present

- Founded and managed Scholarjobs, a platform supporting fresh graduates and job seekers by sharing employment opportunities, scholarship announcements, and career-related resources. **Platform Website.**

Candidate for Ideal student of the year - Faculty of Science, Alexandria university

Jan 2019

Upgrade Alexandria (NGO)

Alexandria, Egypt
2017 - 2018

- Volunteered as an organizer of scientific and soft-skills events and workshops for students.

Al-Furqan Charity Organization (NGO)

Alexandria, Egypt
2013 - 2016

- Participated in the preparation and packaging of free snacks for fasting travelers during the month of Ramadan.
- Participated in many campaigns for cleaning and planting streets.

Honors & Awards

- Champion of the Semester Award**, Alamein International University, (2025–2026 Academic Year) - Awarded in recognition of outstanding research, academic, and administrative achievements.
- Best Faculty Member Award**, Faculty of Science, Alamein International University (2024, 2025).
- Undergraduate Excellence Award**, Faculty of Science, Alexandria University (2019).

References

- Prof. Dr. Mohamed Salem:** Director of Academic Programs, Field of Advanced Basic Sciences, AIU
 - Email: msalem@aiu.edu.eg
- Prof. Sherine Abdel Salam:** Professor of Molecular Biology, Zoology Department, Alexandria university
 - Ph.D. at Bordeaux Segalen University - Email: sherine.abdel.salam@alexu.edu.eg
- Prof. Dr. Marwa Eltoweissy:** Professor of Molecular biology, Zoology Department, Alexandria university
 - Ph.D. at Medical University of Göttingen, Germany - Email: marwaeltoweissy@alexu.edu.eg
- Ass. Prof. Nefertiti El-Nikhely:** Associate Professor of Biochemistry, Alamein International University
 - Ph.D. at Max Planck Institute for Heart & Lung, Germany - Email: nelnikhely@aiu.edu.eg